

Homework 1

(a) $u, v \in \mathbb{R}^{10}$ combine them s.t. $(u_1, v_1, u_2, v_2, \dots, u_{10}, v_{10})$.

1. Create a vector of ones $e = \text{np.ones}(20)$.

2. $e[::2] = u \Rightarrow e = (u_1 \mid u_2 \mid u_3 \mid \dots \mid u_{10} \mid)$

3. $e[1::2] = \overset{\text{odd}}{e[1::2]} \cdot v \Rightarrow e = (u_1 \mid v_1 \mid u_2 \mid v_2 \mid \dots \mid)$

(b) create a random vector $v \in \mathbb{R}^{20}$ and remove those elements which are in $[0.4, 0.6]$.

• Recall the function $\text{np.delete}(v, \text{index})$ can be broken

• Let's obtain indexes: $\text{index} = (u > 0.4) \& (u < 0.6)$
 $\Rightarrow$ boolean array

• np.delete deletes "true" entries, hence we negate it with "~".

(c) Take random $v \in \mathbb{R}^{100}$ and compute

$$v_{\text{new}}[j] = v[j] - v[j-1] \quad j > 0$$

$$v_{\text{new}}[0] = 0.$$

$$v_{\text{new}}[1:] = \underbrace{[v_1, v_2, \dots, v_{99}]}_{v[1:]} - \underbrace{[v_0, v_1, \dots, v_{98}]}_{v[:99]}$$

Finally $v_{\text{new}}[0] = 0.$

Remark: In Python

$$u = \dots$$

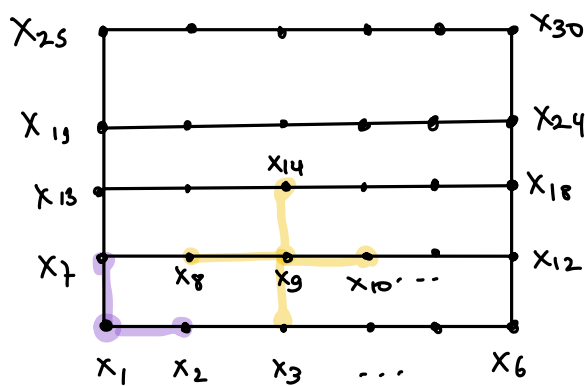
$$v = u$$

does not make a new copy of u .

for that use np.copy .

(d) Take a rectangular grid with 5×6 grid points and a numbering of them $x_1, \dots, x_{30}$. Construct a matrix A

s.t. $A[i,j] = \begin{cases} -4 & x_i = x_j \\ 1 & x_i, x_j \text{ are vertically or horizontally neighbors} \\ 0 & \text{otherwise.} \end{cases}$



↳ what size matrix A has?

$$A \in \mathbb{R}^{30 \times 30}$$

• We know its main diag has -4 s:

$$A = \begin{bmatrix} -4 & & & \\ & -4 & & \\ & & \ddots & \\ & & & -4 \end{bmatrix}$$

That is, when $i=j$

• What property neighbors have in our numbering?

↳ let's look at one row and forget about vertical neighbors : $x_{j-1} - x_j - x_{j+1}$

(only 1st row) $\Rightarrow$

Point	x_1	x_2	x_3	x_4	x_5	x_6
x_1	-4	1	0	0	0	0
x_2	1	-4	1	0	0	0
x_3	0	1	-4	1	0	0
x_4	0	0	1	-4	1	0
x_5	0	0	0	1	-4	1
x_6	0	0	0	0	1	-4

what about vertical neighbors!

We can easily see that for a point x_j , its verticals neighbors are x_{j+6} and x_{j-6} (if possible)

Point	x_1	$\dots$	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}
x_1				1	0	0	0	0	0
x_2				0	1	0	0	0	0
x_3				0	0	1	0	0	0
x_4				0	0	0	1	0	0
x_5				0	0	0	0	1	0
x_6				0	0	0	0	0	1

we know!

;

Identity matrix!

$\Rightarrow$ For the rest row: $\begin{bmatrix} B & I & 0 & 0 & 0 \end{bmatrix}$

This is a tri-diag. matrix with main diag $-(4, 4, 4, \dots, 4)$ off diags $(1, 1, 1, \dots, 1)$.

call this B :

$$B = \begin{bmatrix} -4 & 1 & & \\ 1 & -4 & 1 & \\ & 1 & -4 & 1 \\ & & 1 & -4 \end{bmatrix}$$

Analogously:

For the second row:



Second row has neighbors both in first and third row! where previously we only had vertical neighbors in second row.

all the way to the last row. All together:

$$\begin{bmatrix} B & I & O & O & O \\ I & B & I & O & O \\ O & I & B & I & O \\ O & O & I & B & I \\ O & O & O & I & B \end{bmatrix}$$

This block matrix is made of three block diagonal matrices!